edited. Instead if you select one of these, you will have the option to **Copy** the locked dataset and then you can edit the copied dataset.)

When you have finished loading your monitor data, click **OK** in the **Manage Monitor Datasets** window. This will take you back to the **Modify Datasets** window, which will show the name of the **Dataset**(s) that you just entered.

4.1.3.2 Format for Monitor Data

Monitor data is required to be formatted in a single database file, with monitor definition information and monitor values in a single line. Tables 4-6a and 4-6b list the variables in the monitor dataset and provide a sample of what a data file might look like.

NOTE: The monitor data files do not specify the pollutant with which the data is associated—this is specified by the user when loading the monitor data into BenMAP-CE.

Table 4-6a. Required Format, Air Monitoring Data File Variables

Variable	Type	Required	Notes
Monitor Name	Text	Yes	Unique name for each monitor in a particular location.
Monitor Description	Text	No	Description of the Monitor.
Longitude	Numeric (double)	Yes	Values should be in decimal degree format. Values in the eastern hemisphere are positive, and those in the western hemisphere are negative.
Latitude	Numeric (double)	Yes	Values should be in decimal degree format. Values in the northern hemisphere are positive, and those in the southern hemisphere are negative.
Metric	Text	No	This variable is either blank (signifying that the Values are Observations, rather than Metric values), or must reference an already defined Metric (e.g., 1-hour daily maximum) for the appropriate Pollutant.
Seasonal Metric	Text	No	This variable is either blank (signifying that the Values are not Seasonal Metric values) or must reference an already defined Seasonal Metric for the Metric (e.g., mean of the 1-hour maximum values for the months of June through August).
Statistic	Text	No	This is an annual metric, which is either blank (signifying that the values are not annual statistics) or must be one of: None, Mean, Median, Max, Min, Sum. (e.g., mean of the 1-hour maximum for the year)
Values	Text	Yes	If Metric is blank, values are supplied as a comma-delimited string of values for the year [e.g., 365 or 366 (leap year) values for daily data, 8760 or 8784 (leap year) values for hourly data]. If Metric is defined, but Seasonal Metric and Statistic are blank, 365 or 366 metric values. If Seasonal Metric is defined, but Statistic is blank, n seasonal metric values. If Statistic is defined, one annual statistic value for either the Metric (if Seasonal Metric is blank) or the Seasonal Metric. Missing values are signified with a period (‘.’).

Table 4-6b. Required Format, Sample Air Monitoring Data File

Monitor Name	Monitor Description	Latitude	Longitude	Metric	Seasonal Metric	Statistic	Values
260050003881011	'RESIDENTIAL, SUBURBAN'	42.7678	-86.1486	D24HourMean	QuarterlyMean	Mean	11.67
260170014881011	'RESIDENTIAL, URBAN AND CENTER CITY'	43.5714	-83.8907	D24HourMean	QuarterlyMean	Mean	11.62
260210014881011	'COMMERCIAL, RURAL'	42.1978	-86.3097	D24HourMean	QuarterlyMean	Mean	11.58
260490021881011	'RESIDENTIAL, URBAN AND CENTER CITY'	43.0472	-83.6702	D24HourMean	QuarterlyMean	Mean	11.56
260650012881011	'RESIDENTIAL, URBAN AND CENTER CITY'	42.7386	-84.5346	D24HourMean	QuarterlyMean	Mean	11.49
260770008881011	'COMMERCIAL, URBAN AND CENTER CITY'	42.2781	-85.5419	D24HourMean	QuarterlyMean	Mean	11.44
260810007881011	'INDUSTRIAL, URBAN AND CENTER CITY'	42.9561	-85.6791	D24HourMean	QuarterlyMean	Mean	11.4
260810020881011	'INDUSTRIAL, URBAN AND CENTER CITY'	42.9842	-85.6713	D24HourMean	QuarterlyMean	Mean	11.36
260990009881011	'COMMERCIAL, SUBURBAN'	42.7314	-82.7935	D24HourMean	QuarterlyMean	Mean	11.43
261010922881011	'RESIDENTIAL, URBAN AND CENTER CITY'	44.3070	-86.2426	D24HourMean	QuarterlyMean		11.43,11.07,10.98,10.90
261130001881011	'FOREST, RURAL'	44.3106	-84.8919	D24HourMean	QuarterlyMean		11.44,14.23,11.20,9.30
261150005881011	'AGRICULTURAL, RURAL'	41.7639	-83.4719	D24HourMean	QuarterlyMean		11.39,13.71,10.04,11.24
261210040881011	'COMMERCIAL, URBAN AND CENTER CITY'	43.2331	-86.2386	D24HourMean	QuarterlyMean		11.28,9.23,15.03,11.10
261250001881011	'RESIDENTIAL, SUBURBAN'	42.4631	-83.1832	D24HourMean	QuarterlyMean		11.27,11.12,13.15,12.03
261390005881011	'RESIDENTIAL, SUBURBAN'	42.8945	-85.8527	D24HourMean	QuarterlyMean		11.27,10.05,10.9,12.13
261470005881011	'RESIDENTIAL, SUBURBAN'	42.9533	-82.4562	D24HourMean	QuarterlyMean		11.26,12.13,15.00,10.01
261610008881011	'COMMERCIAL, URBAN AND CENTER CITY'	42.2406	-83.5996	D24HourMean	QuarterlyMean		11.24,14.05,11.03,10.10
261630001881011	'COMMERCIAL, SUBURBAN'	42.2286	-83.2082	D24HourMean	QuarterlyMean		11.23,9.23,13.04,10.09
261630015881011	'COMMERCIAL, URBAN AND CENTER CITY'	42.3028	-83.1065	D24HourMean	QuarterlyMean		11.23,11.24,10.45,10.04
261630016881011	'RESIDENTIAL, URBAN AND CENTER CITY'	42.3578	-83.0960	D24HourMean	QuarterlyMean		11.2,10.73,11.23,10.45
261630019881011	'RESIDENTIAL, SUBURBAN'	42.4308	-83.0001	D24HourMean			.,.,13.01,.,.,17.19,.,.,3.89,.,.,13.86,.,.,5.9
261630025881011	'COMMERCIAL, SUBURBAN'	42.4231	-83.4263	D24HourMean			.,.,13.48,.,.,8.73,.,.,6.079,.,.,14.534,.,.,3.
261630033881011	'INDUSTRIAL, SUBURBAN'	42.3067	-83.1488	D24HourMean			.,.,15.675,.,.,12.72,.,.,6.46,.,.,16.24,.,.,6.
261630036881011	'COMMERCIAL, SUBURBAN'	42.1873	-83.1539	D24HourMean			.,.,16.52,.,.,15.01,.,.,5.41,.,.,15.39,.,.,5.0
261630038881011	'RESIDENTIAL, URBAN AND CENTER CITY'	42.3350	-83.1096	D24HourMean			.,.,20.52,.,.,21.85,.,.,5.88,.,.,19.28,.,.,5.1
261630039881011	'RESIDENTIAL, URBAN AND CENTER CITY'	42.3233	-83.0685	D24HourMean			.,.,20.14,.,.,7.59,.,.,8.17,.,.,13.11,.,.,7.59

4.1.4 Incidence and Prevalence Rates Data

Most health impact functions, such as those developed from log-linear or logistic health impact functions, estimate the percent change in a health effect associated with a pollutant change. In order to estimate the absolute change in incidence using these functions, the baseline incidence rates (and in some cases the prevalence rate) of the adverse health effect are needed.

The incidence rate is the number of health effects per person in the population per unit of time, and the prevalence rate is the percentage of people that suffer from a particular chronic illness. For example, the incidence rate for asthma attacks may be 25 cases per asthmatic individual per year, and the prevalence rate (measuring the percentage of the population that is asthmatic) might be six percent of the total population.

Fundamental Concepts – Incidence and Prevalence

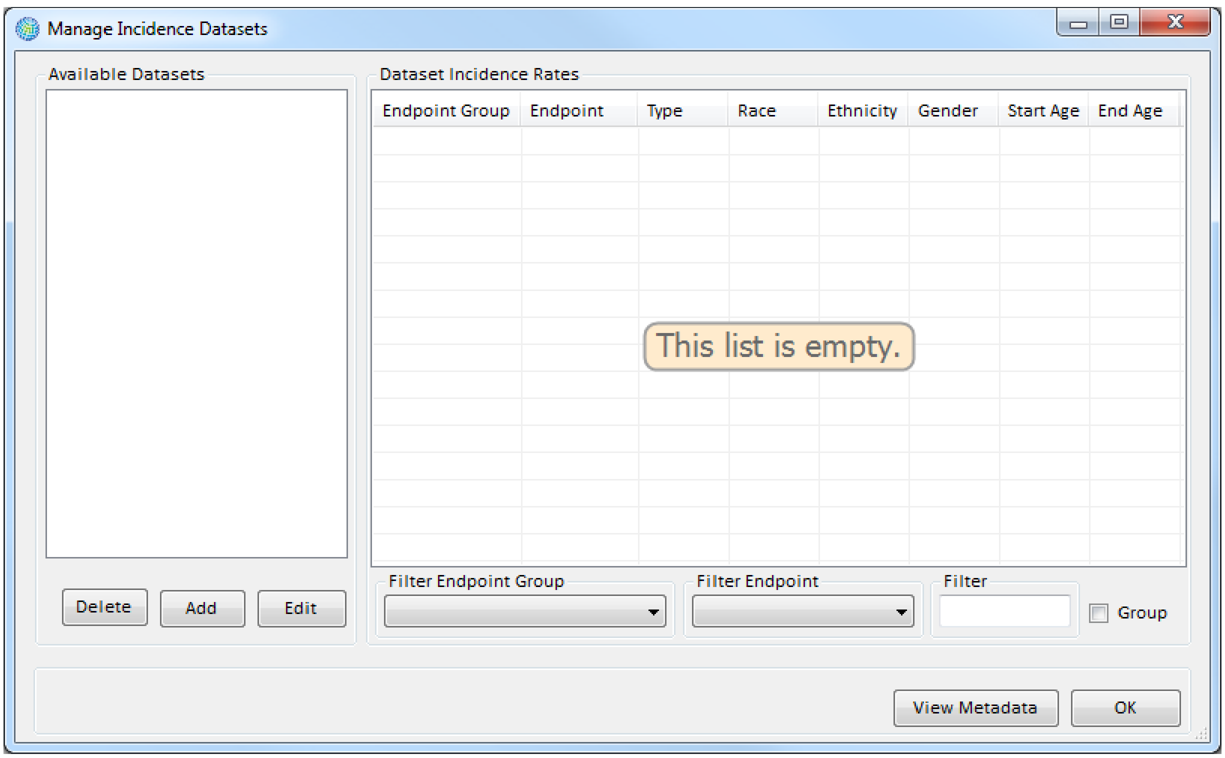
Incidence is a measure of the total number of new occurrences of an adverse health impact in a geographic area over time. The **incidence rate** is the average number of adverse health effects (e.g., respiratory hospital admissions) per person per unit of time, typically a day or a year. The **incidence rate** must be expressed at the same time scale as the specified by the health impact function. For example, a health impact function quantifying day-to-day changes in premature death requires a daily mortality rate. The **baseline incidence rate**, also called the **background incidence rate,** is the incidence of a given adverse effect due to all causes including air pollution. BenMAP typically estimates and reports benefits as the change in incidence between the Baseline and Control scenarios, (e.g., the number of avoided asthma Emergency Department visits).

The **prevalence rate** is the percentage of individuals in a given population at a given point in time who are experiencing or have been diagnosed with a given adverse health condition (e.g., the prevalence of asthmatics among children 0 – 17). It may be required for certain health impact functions, such as those that focus on asthmatics or other groups exhibiting a co-morbidity.

NOTE: For both incidence and prevalence rates, BenMAP-CE allows the user to have rates that vary by race, ethnicity, gender, and age group. BenMAP-CE can support multiple sets of incidence and prevalence rates, if the rates differ by year or by grid definition.

4.1.4.1 Add Incidence/Prevalence Rates

To start adding baseline incidence and prevalence data files, click on the **Manage** button below the **Incidence/Prevalence Rates** box. The **Manage Incidence Datasets** window will appear.



In this window you may **Add**, **Edit**, and **Delete** datasets. The section on the left under **Available Datasets** lists the incidence/prevalence datasets that are currently in the setup. The section on the right under the **Dataset Incidence Rates** identifies the rates in the selected dataset.

To add a dataset, click the **Add** button. This will bring up the **Incidence Dataset Definition** window. Give a name to the dataset that you are creating by typing a name in **Dataset Name** box.